```
import pandas as pd

df = pd.read_csv("/content/Traffic Accident Dataset Process.csv")

print(df.head())
print(df.info())
```

```
   Accident_ID  Date  Time  Location  Weather  Road_Type  Vehicle_Type  \
0            0     0  1338         3        3          2             3
1         1111     1  1418         0        2          0             1
2         2222     2   755         3        3          2             0
3         3333     3   264         3        0          1             1
4         4444     4    52         1        2          0             3

   Driver_Age  Severity  Casualties  Speed_Limit
0          37         1           2           60
1          33         2           2           60
2          63         1           4           40
3          56         2           0           40
4          66         0           3          100
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9131 entries, 0 to 9130
Data columns (total 11 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Accident_ID   9131 non-null   int64
 1   Date          9131 non-null   int64
 2   Time          9131 non-null   int64
 3   Location      9131 non-null   int64
 4   Weather       9131 non-null   int64
 5   Road_Type     9131 non-null   int64
 6   Vehicle_Type  9131 non-null   int64
 7   Driver_Age    9131 non-null   int64
 8   Severity      9131 non-null   int64
 9   Casualties    9131 non-null   int64
 10  Speed_Limit   9131 non-null   int64
dtypes: int64(11)
memory usage: 784.8 KB
None
```

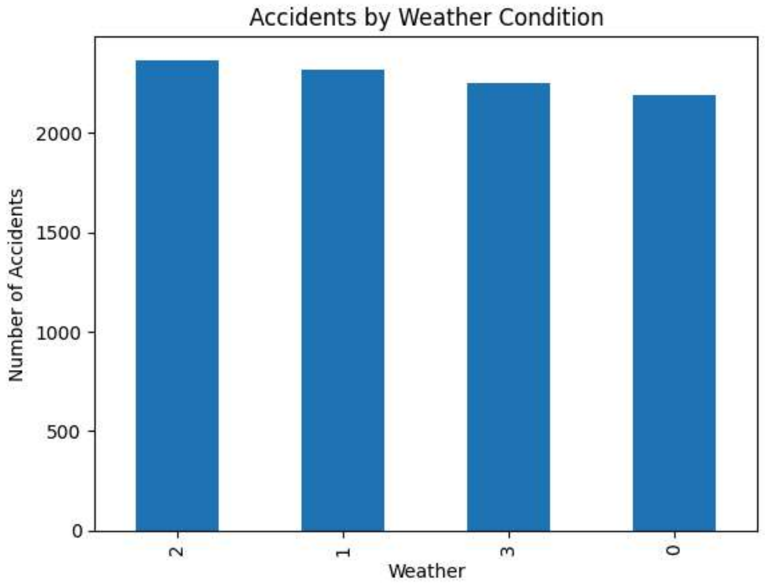
```
print(df.isnull().sum())
```

```
Accident_ID     0
Date            0
Time            0
Location        0
Weather         0
Road_Type       0
Vehicle_Type    0
Driver_Age      0
Severity        0
Casualties      0
Speed_Limit     0
dtype: int64
```

```
df = df.dropna()
```

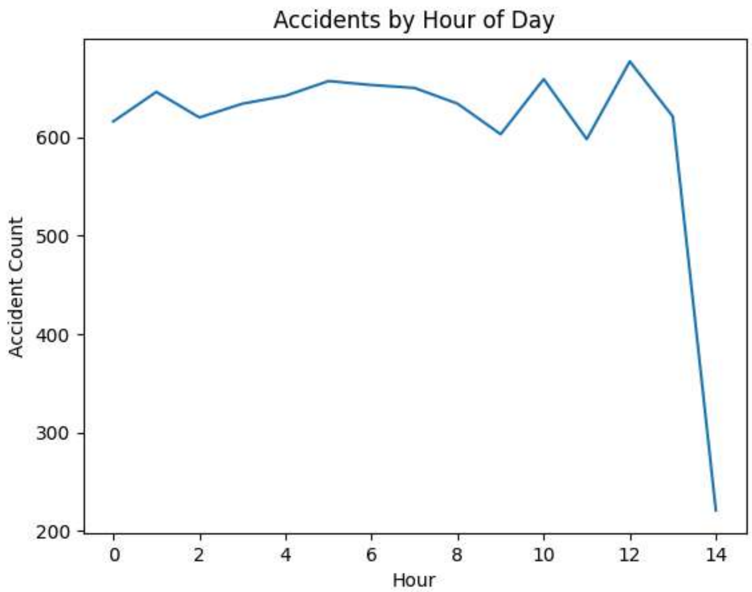
```
df.drop_duplicates(inplace=True)
```

```
import matplotlib.pyplot as plt

weather_counts = df['Weather'].value_counts().head(10)

weather_counts.plot(kind='bar')
plt.title("Accidents by Weather Condition")
plt.xlabel("Weather")
plt.ylabel("Number of Accidents")
plt.show()
```

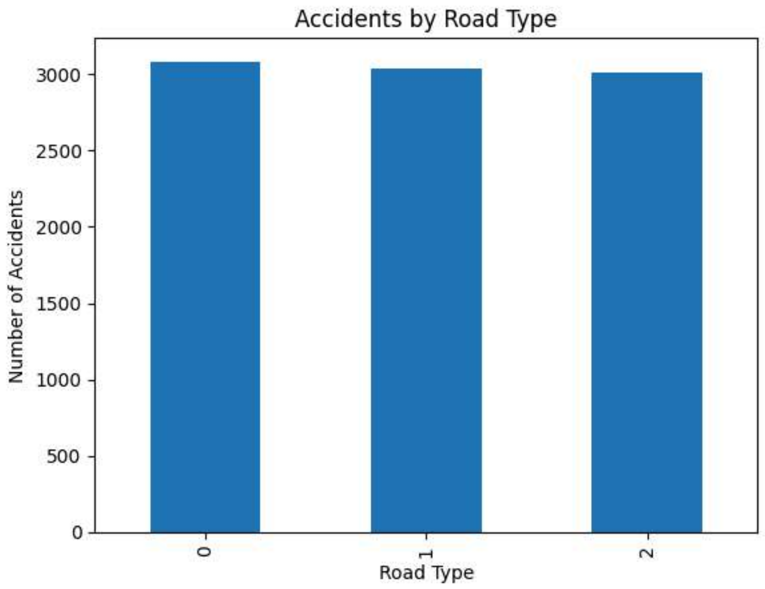



```
df['Hour'] = df['Time'] // 100
```

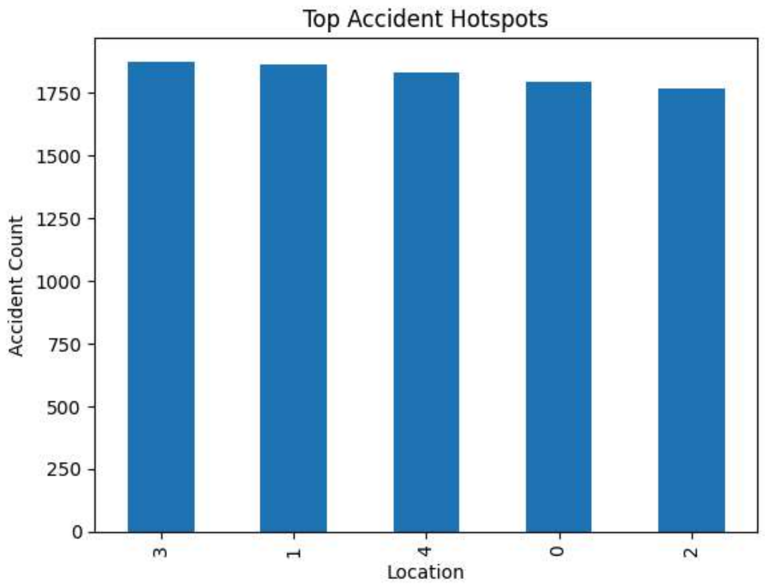
```
df['Hour'] = df['Time'] // 100
hour_counts = df['Hour'].value_counts().sort_index()

hour_counts.plot(kind='line')
plt.title("Accidents by Hour of Day")
plt.xlabel("Hour")
plt.ylabel("Accident Count")
plt.show()
```




```
road_counts = df['Road_Type'].value_counts()

road_counts.plot(kind='bar')
plt.title("Accidents by Road Type")
plt.xlabel("Road Type")
plt.ylabel("Number of Accidents")
plt.show()
```




```
city_counts = df['Location'].value_counts().head(10)

city_counts.plot(kind='bar')
plt.title("Top Accident Hotspots")
plt.xlabel("Location")
plt.ylabel("Accident Count")
plt.show()
```



Start coding or generate with AI.